Kernel Method

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1 Preliminaries

Vector Space: A *vector space* $\mathcal{V}$ is a set with all its elements satisfying the axioms of vector operations. That is, for any $x, y, z \in \mathcal{V}$ and constants $\alpha, \beta \in \mathbb{R}$, it has

- $(x + y) + z = x + (y + z)$, $x + y = y + x$.
- There exists a zero vector $0 \in \mathcal{V}$ such that $x + 0 = x$.
- For each $x \in \mathcal{V}$, there exists a unique vector $-x \in \mathcal{V}$ such that $x + (-x) = 0$.
- $1x = x$, $\alpha(\beta x) = (\alpha\beta)x$, $\alpha(x + y) = \alpha x + \alpha y$,$(\alpha + \beta)x = \alpha x + \beta x$.

The vector space concerns the *structure* rather than the *representation* of the elements. The elements in a set can be anything, including numbers, vectors and functions. As long as these elements follow the structure defined by the axioms above, we call the set a vector space.

In particular, a vector space with elements being functions is called a *function space*. Let $\mathcal{C}$ be a function space of all functions $X \to \mathbb{R}$, then for any $f, g \in \mathcal{C}$, $f + g$ and αf are also in $\mathcal{C}$, where

- $\forall x \in X, (f + g)(x) = f(x) + g(x)$
- $\forall x \in X, (\alpha f)(x) = \alpha f$

Normed Vector Space: If $\mathcal{V}$ has a norm $\|\cdot\| : \mathcal{V} \to \mathbb{R}$ defined on it, we say $\mathcal{V}$ is a *normed vector space*. The norm satisfies that for any $x, y \in \mathcal{V}$ and constant $\alpha \in \mathbb{R}$,

- $\|x + y\| \leq \|x\| + \|y\|$.
- $\|\alpha x\| = |\alpha|\|x\|$.
- $\|x\| = 0$ only when $x = 0$.

Linear Transformation and Linear Functional: Let $\mathcal{V}$ and $\mathcal{V}_1$ be two vector spaces. A *linear transformation* is a mapping $\mathcal{T} : \mathcal{V} \to \mathcal{V}_1$ satisfying that for any $x, y \in \mathcal{V}$ and constants $\alpha, \beta \in \mathbb{R}$,

$$\mathcal{T}(\alpha x + \beta y) = \alpha\mathcal{T}(x) + \beta\mathcal{T}(y)$$

If $\mathcal{V}_1 = \mathbb{R}$, we say $\mathcal{T}$ is a *linear functional*, denoted as T.

If $\mathcal{V}$ is a normed vector space, then we say T is *bounded* if there exists a constant $C > 0$ such that $T(x) \leq C\|x\|$ for any $x \in \mathcal{V}$. The space of all bounded linear functionals on $\mathcal{V}$ is called the *dual space* of $\mathcal{V}$, denoted as $\mathcal{V}^*$.

Inner Product: Let $\mathcal{X}$ be a vector space, an inner product is a function $\langle\cdot,\cdot\rangle : \mathcal{X}\times\mathcal{X} \to \mathbb{R}$, which satisfies that for any $x, y \in \mathcal{X}$ and $\alpha, \beta \in \mathbb{R}$,

- $\langle x, y\rangle = \langle y, x\rangle$.
- $\langle \alpha x + \beta y, z\rangle = \alpha\langle x, z\rangle + \beta\langle y, z\rangle$.
- $\langle x, x\rangle > 0$ if $x \neq 0$.

For example, if $\mathcal{X} = \mathbb{R}^k$, then for any $x, y \in \mathcal{X}$, given any $k \times k$ symmetric positive-definite matrix M, we can define the inner product as $\langle x, y\rangle = x^T M y$. If $M = I$, then $\langle x, y\rangle = x^T y$, which is called the dot product.

Let $\mathcal{F}$ be the set of all continuous functions mapping from $[0, 1]$ to $\mathbb{R}$. For any $f, g \in \mathcal{F}$, we can define the inner product as a Riemann integral, i.e., $\langle f, g\rangle = \int_0^1 f(t)g(t)\ dt$. In this case, both f and g can be considered as infinite-dimensional vectors with entries in $\mathbb{R}$ and real indexes in $[0, 1]$, and $\langle f, g\rangle$ resembles the dot product.

Hilbert Space: A Hilbert space $\mathcal{H}$ is a vector space with inner product $\langle \cdot, \cdot\rangle_{\mathcal{H}} : \mathcal{H} \times \mathcal{H} \to \mathbb{R}$ defined on it. Since an inner product induces a norm by defining $\|x\| := \sqrt{\langle x, x\rangle_{\mathcal{H}}}$, a Hilbert space is also a normed vector space.

Let $(X, \mathcal{M}, \mu)$ be a measure space where X is a set, $\mathcal{M}$ is a σ-algebra of X and μ is a corresponding measure defined on $\mathcal{M}$. For any measurable function $f : X \to \mathbb{R}$, define the norm as the Lebesgue integral $\|f\| = \left(\int |f|^2\ d\mu\right)^{1/2}$, we say all f with $\|f\| < \infty$ form a normed vector space $L^2(\mu)$.

$L^2(\mu)$ can be a Hilbert space by defining the inner product as

$$\langle f, g\rangle_{\mathcal{H}} = \int fg\ d\mu$$

Since Lebesgue integral has properties $\int (f + g)\ d\mu = \int f\ d\mu + \int g\ d\mu$ and $\int \alpha f\ d\mu = \alpha \int f\ d\mu$, it is easy to check that this inner product definition is valid.

Let A be some index set, and let $\{u_\alpha\}_{\alpha \in A}$ be a subset of elements (possibly uncountablely many) in $\mathcal{H}$. We say $\{u_\alpha\}_{\alpha \in A}$ is *orthonormal* if $\|u_\alpha\| = 1$ for all $\alpha \in A$ and $\langle u_\alpha, u_\beta\rangle_{\mathcal{H}} = 0$ for any $\alpha \neq \beta, \alpha, \beta \in A$. If, in further, any $x \in A$ can be expressed as $x = \sum_{\alpha \in A} \langle x, u_\alpha\rangle_{\mathcal{H}} u_\alpha$ (i.e., $\{u_\alpha\}_{\alpha \in A}$ has the largest possible cardinality), we say $\{u_\alpha\}_{\alpha \in A}$ is an *orthonormal basis* of $\mathcal{H}$ (Theorem 5.27, [2]; Theorem 4.18, [1]), and its cardinality represents the number of dimensions of $\mathcal{H}$.

We say $\mathcal{H}$ is *separable* if and only if its orthogonal basis is countable (Theorem 5.29, [2]). $L^2(\mu)$ is separable if $X = \mathbb{R}^n$ and μ is a Lebesgue measure (Exercise 62, [2]).

Riesz Representation Theorem: It is well known that if $\mathcal{T} : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation, then there must exist a unique matrix $A \in \mathbb{R}^{m \times n}$ such that $\mathcal{T}(x) = Ax$ for any $x \in \mathbb{R}^n$. That is to say, $\mathcal{T}$ is represented by A, and the transformation property

$$\mathcal{T}(\alpha x + \beta y) = \alpha \mathcal{T}(x) + \beta \mathcal{T}(y) \tag{1}$$

can be equivalently written as

$$A(\alpha x + \beta y) = \alpha Ax + \beta Ay \tag{2}$$

Here, $\mathcal{T}$ is an *general function*: we only know the input and output of $\mathcal{T}$ satisfy the linear relationship defined in Eq (1), but we may not have an explicit formulation of $\mathcal{T}$. Eq (2) means in the Euclidean space case, each $\mathcal{T}$ can be *one-to-one mapped to* an explicit function A in the space $\mathbb{R}^{m \times n}$, called the *representation* of $\mathcal{T}$, such that Ax always equals to $\mathcal{T}(x)$ for any input x.

The special case of $m = 1$ can be stated as follows: for any bounded linear functional $T : \mathbb{R}^n \to \mathbb{R}$, there exists a unique vector $a \in \mathbb{R}^n$ such that $T(x) = \langle a, x\rangle = a^T x$ for any $x \in \mathbb{R}^n$. Note that both a and x come from the same space $\mathbb{R}^n$.

More generally, Riesz Representation Theorem (Theorem 4.12 in [1]; Theorem 5.25 in [2]) states that, given a Hilbert space $\mathcal{H}$, for any linear functional $T \in \mathcal{H}^*$, there exists a unique $x \in \mathcal{H}$ such that $T(x) = \langle x, y \rangle_{\mathcal{H}}$ for any $y \in \mathcal{H}$. [1]

Gram Matrix: Given $v_1, v_2, ...v_n \in \mathcal{H}$, a *Gram matrix* is defined as $G \in \mathbb{R}^{n \times n}$ where $G_{ij} = \langle v_i, v_j \rangle_{\mathcal{H}}$ for $i, j \in \{1, ..., n\}$. If $\mathcal{H}$ is the Euclidean space $\mathbb{R}^m$ with inner product defined as vector dot product, then we can decompose it as $G = A^T A$ where $A = [v_1, v_2, ..., v_n]$.

A Gram matrix is always positive semi-definite (Theorem 7.2.10., [3]): For any $x \in \mathbb{R}^n$, we have

$$x^T G x = \sum_{i=1}^{n} \sum_{j=1}^{n} \langle v_i, v_j \rangle_{\mathcal{H}} x_i x_j = \sum_{i=1}^{n} \sum_{j=1}^{n} \langle x_i v_i, x_j v_j \rangle_{\mathcal{H}} = \left\langle \sum_{i=1}^{n} x_i v_i, \sum_{j=1}^{n} x_j v_j \right\rangle_{\mathcal{H}} = \left\| \sum_{i=1}^{n} x_i v_i \right\| \geq 0$$

2 Kernel Method

Kernel: Let X be a vector space, a kernel is a function $k(\cdot, \cdot) : X \times X \to \mathbb{R}$ satisfying two properties:

- **symmetric**: For any $x, y \in X$, $k(x, y) = k(y, x)$.
- **positive semi-definite** [2]: For any $n \in \mathbb{Z}^+$ and $x_1, x_2, ..., x_n \in X$, define the matrix $M \in \mathbb{R}^{n \times n}$ by $M_{ij} = k(x_i, x_j)$, then M is always positive semi-definite, i.e., for any $c_1, ..., c_n \in \mathbb{R}$,
$$\sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j k(x_i, x_j) \geq 0$$

Mercer's theorem (Theorem 2.1, [5]) shows that, if k is continuous and positive semi-definite, then there exists a mapping $\phi(\cdot) : X \to L^2(\mu)$ such that k can be represented by an inner product form: for any $x, y \in X$,

$$k(x, y) = \langle \phi(x), \phi(y) \rangle_{L^2(\mu)}$$

This can be expressed as the following commutative diagram (taking $\mathcal{H} = L^2(\mu)$) [6].

$$\begin{tikzcd}
X \times X \arrow[d, "\phi"'] \arrow[dr, "k"] & \\
\mathcal{H} \times \mathcal{H} \arrow[r, "\langle \cdot, \cdot \rangle_{\mathcal{H}}"'] & \mathbb{R}
\end{tikzcd}$$

Mercer's Theorem [5, 6, 7]: *Suppose $X \in \mathbb{R}^n$ is compact and μ is the Lebesgue measure. Construct a mapping $T_k : L^2(\mu) \to L^2(\mu)$ such that $T_k(f) = \int k(x, \cdot) f(x) dx$ (here $k(x, \cdot)$ means $k(\cdot, \cdot)$ with the first parameter fixed to be x). If k is positive semi-definite, i.e., for any $f \in L^2(\mu)$,*

$$\langle f, T_k(f) \rangle = \int f(y) \left(\int k(x, y) f(x) \, dx \right) \, dy = \int k(x, y) f(x) f(y) \, dx dy \geq 0$$

[1] In some textbooks like [1, 2], Riesz representation theorem is referred to a different theorem called Riesz–Markov–Kakutani representation theorem, which states that for any bounded linear functional T on $C_c(X)$ (or $C_0(X)$), there exists a Radon measure μ such that $T(f) = \int f \, d\mu$ for any f in $C_c(X)$ (or $C_0(X)$). This theorem is related to Radon measure rather than Hilbert space.

[2] Some literature like [5] call this condition positive definite, but more rigorously speaking, it should be called semi-definite.

Then, suppose $\{u_i\}_{i=1}^{\infty}$ is the orthonormal basis of $L^2(\mu)$, there exist fixed non-negative constants $\{\lambda_i\}_{i=1}^{\infty}$ such that for any $x, y \in X$,

$$k(x,y) = \sum_{i=1}^{\infty} \lambda_i u_i(x) u_i(y) \tag{3}$$

And the converse is also true.

By Eq (3), defining $\phi(x) = \sum_{i=1}^{\infty} \sqrt{\lambda_i} u_i(x) u_i$, then $k(x,y) = \langle \phi(x), \phi(y) \rangle$.

Reproducing Kernel Hilbert Space: Let $\mathcal{H}$ be the Hilbert space of the functions $f : X \to \mathbb{R}$. Fixing any $x \in X$, we can construct a map from f to $f(x)$ for any $f \in \mathcal{H}$. Denote such map as T_x, then for any $f, g \in \mathcal{H}$ and $\alpha, \beta \in \mathbb{R}$, we have

$$T_x(\alpha f + \beta g) = (\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x) = \alpha T_x(f) + \beta T_x(g)$$

implying that T_x is a linear functional. We call T_x the *evaluation functional* for the point x.

By Riesz representation theorem, for each T_x, there exists a unique $k_x \in \mathcal{H}$ such that

$$T_x(f) = f(x) = \langle f, k_x \rangle_{\mathcal{H}} \quad \text{for any } f \in \mathcal{H} \tag{4}$$

Note that the function $k_x : X \to \mathbb{R}$ depends on x. Define $\mathcal{H}' = \{k_x : x \in X\} \subset \mathcal{H}$ as all such k_x that represents a evaluation functional, then for any $x, y \in X$, we have

$$k_x(y) = T_y(k_x) = \langle k_x, k_y \rangle_{\mathcal{H}} = \langle k_y, k_x \rangle_{\mathcal{H}} = T_x(k_y) = k_y(x) \tag{5}$$

Eq (5) shows that we can define a kernel $k(x,y) = \langle \phi(x), \phi(y) \rangle_{\mathcal{H}}$ where the mapping ϕ is defined as $\phi(x) = k_x$. We call such k a *reproducing kernel* since its partial application k_x reproduces the input by Eq (4) [3]. Any Hilbert space containing a reproducing kernel is called a *Reproducing Kernel Hilbert Space (RKHS)*.

RKHS is a powerful analysis tool in machine learning. Let X be the space of inputs where each $x \in X$ is a sample. $\mathcal{H}$ be the space where each $f \in \mathcal{H}$ is a model. Given any x, a model f produces an output $f(x)$. However, f can be a non-linear function like a neural network, which is typically difficult to analyze. Eq (4) gives an interesting insight: Every $x \in X$ can be mapped to a (possibly higher-dimensional) representer k_x in $\mathcal{H}$, called the *kernel embedding*. Hence, for any model $f \in \mathcal{H}$, the non-linear evaluation $f(x)$ can always be converted into a linear evaluation $\langle f, k_x \rangle_{\mathcal{H}}$.

Kernel Mean Embedding [5]: We have introduced how each point $x \in X$ can be mapped to an embedding k_x in $\mathcal{H}$. In general, suppose $x \in X$ follows some distribution P, which is a probability measure defined on the measurable space (X, Σ, P) (Note that P is different from the measure μ defined on $(X, \mathcal{M}, \mu)$), we can map P to a kernel embedding in $\mathcal{H}$ by

$$\phi(P) = \int k_x \, dP = \mathbb{E}_{x \sim P}[k_x] \tag{6}$$

This mapping is valid as long as P satisfies $\mathbb{E}_{x \sim P} \|k_x\| < \infty$. The right hand side of Eq (6) shows that the embedding is in fact the expectation of kernels over P, so we call it *kernel mean embedding*.

For any model f, the kernel mean embedding allows us to evaluate $\mathbb{E}_{x \sim P}[f(x)]$ by the inner product $\langle f, \mathbb{E}_{x \sim P}[k_x] \rangle_{\mathcal{H}}$, i.e.,

$$\mathbb{E}_{x \sim P}[f(x)] = \int f \, dP = \int \langle f, k_x \rangle_{\mathcal{H}} \, dP = \int \left(\int f k_x \, d\mu \right) dP = \int f \left(\int k_x \, dP \right) d\mu = \langle f, \mathbb{E}_{x \sim P}[k_x] \rangle_{\mathcal{H}}$$

[3] Reproducing kernel is a special kernel with reproducing property; not all kernel has reproducing property.

The above equation utilizes Fubini–Tonelli theorem. In the third term, we write $f = f(y)$ and $k_x = k(x, y)$. Note that μ is for y and P is for x.

References

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Appendix

Counting measure and Lebesgue measure: Let us consider the classic inner product in $\mathbb{R}^n$. Suppose $n = 3$, let $v_1 = [1, 2, 3]^T$ and $v_2 = [5, 6, 4]^T$, we have $\langle v_1, v_2 \rangle = 1 \cdot 5 + 2 \cdot 6 + 3 \cdot 4$. This can be written as an Lebesgue integral with counting measure.

Let $A = \{a_1, a_2, a_3\}$ be a finite set, c be a counting measure on A such that $c(a_i) = 1$ for $i = 1, 2, 3$. Define $v_1 : A \to \mathbb{R}$ such that $v_1(a_1) = 1, v_1(a_2) = 2, v_1(a_3) = 3$. Define $v_2 : A \to \mathbb{R}$ such that $v_2(a_1) = 5, v_1(a_2) = 6, v_1(a_3) = 4$. In this case, $(v_1 v_2)^{-1}(t) \in A$ only when $t = 5, 12$, and we can express the inner product as

$$\int_A v_1 v_2 \, dc = 5 \cdot c((v_1 v_2)^{-1}(5)) + 12 \cdot c((v_1 v_2)^{-1}(12)) = 5 \cdot (c(a_1)) + 12 \cdot (c(a_2) + c(a_3)) = 29$$

Now we suppose A is a continuous interval $[0, 1]$, and define continuous functions $v_1(x) = x, v_2(x) = x$ for any $x \in A$. In this case, $v_1 v_2(x) = x^2 \in [0, 1]$. It is clear that the image of $v_1 v_2$ contains infinite many values, and the counting measure c will result $\int_A v_1 v_2 \, dc = \infty$. In this case, we use the Lebesgue measure μ instead of c. For any interval $[a, b] \in \mathbb{R}(b > a)$, its Lebesgue measure is defined as $\mu([a, b]) = b - a$ (also for $[a, b), (a, b]$ or (a, b)). For any set $Q \in \mathbb{R}$ with countably many elements, $\mu(Q) = 0$. Under μ, the integral $\int_A v_1 v_2 \, d\mu < \infty$.

In contrast, in the earlier case when A is defined as a discrete set and v_1, v_2 are defined as discrete functions, $\int_A v_1 v_2 \, d\mu = 0$.

Note that for any set S, with finite, countably many or uncountably many elements, we have $c(S) \geq \mu(S)$. When switching from discrete sum to continuous integral, we in fact switch from counting measure to Lebesgue measure, which is "smaller" and avoid the integral goes to infinity.